

Well-to-Surface Digital Twin: Operational Summary Report

Asset: Baghewala Field Heavy Oil Operations

Automated SCADA Orchestrator Engine

2026-09-06 18:37:37

1 Asset & Executive Status

Well Identifier:	BGW-JH-07	Crude Gravity:	18.0° API[cite: 7, 8]
True Vertical Depth:	1,200 m[cite: 6]	Operational State:	Normal Operation
Production Elapsed:	16.5 Days	Cycle Cutoff Status:	STABLE PRODUCTION

2 Coupled Thermodynamic & Viscosity State

Metric	Current Value	Baseline Floor	Units
Reservoir Temperature $T(t)$	113.01	46.00	°C[cite: 2, 7]
Crude Viscosity $\mu(t)$	2.21	66.00	cP[cite: 3, 7]
Safe Pumping Limit SPM_{safe}	8.00	2.00	SPM[cite: 5, 8]
Operating Pump Speed	5.00	—	SPM
VFD Operational Frequency	30.0	16.70	Hz

3 Mechanical Dynamics & Dyno Card Diagnostics

Kinematic Parameter	Magnitude	Design Safe Limit
Peak Polished Rod Load (PPRL)	20092.6 lbf	24,000.0 lbf
Estimated Rod Elastic Stretch (ΔL)	1.88 in	18.00 in
Morphology Classification	Normal Operation	Full Rectangular Envelope

4 Lifecycle Economics & Cycle Cutoff Criteria

The joint economic engine balances gross crude revenue against thermal injection costs and electrical wear:

$$J = \int_0^t [P_o q_o(\tau) - C_e P_m(\tau) - C_w \sigma_r(\tau)] d\tau - C_s m_s[\text{cite : 2, 5}]$$

- **Cumulative Net Profit (J):** \$124,500.00
- **Marginal Profit Rate (dJ/dt):** \$1791.56 / day[cite: 2, 7]
- **Cumulative Steam-Oil Ratio (SOR):** 3.18 m³/m³
- **Autonomous Action:** Continue automated VFD throttle modulation.

Report generated deterministically by the SIH 2026 Well2Surface Twin Pipeline.